

Deterministic plan

Stochastic plan

Phase A — Planning (each plan selects facilities z , capacities C , and assignments x before any uncertainty is revealed)

Patient cohort (identical in both columns) — Tier H ($\times 10$) • red, Tier M ($\times 25$) • amber, Tier L ($\times 15$) • green

Row 1

Row 2

First-stage plan — facility selection z and capacity C

Different first-stage decisions: same out-of-sample world.

Phase B — Evaluation (frozen plans meet the same stochastic world)

Scenario ω — same realization in both columns (red X = patient failed at assigned facility: det 12 failures, SP 5 failures)

Row 4

Row 5

Recourse decisions — check = treated, R = re-manufactured, X = cancelled

Deterministic plan — scenario outcome

Treated: 38 Re-mfg: 7 Cancelled: 5 (Tier-H cancelled: 4/10)

Stochastic plan — scenario outcome

Treated: 45 Re-mfg: 4 Cancelled: 1 (Tier-H cancelled: 0/10)

Patient tiers:

Tier H — high urgency ($\times 10$, $\beta=0.55$) Tier M — medium urgency ($\times 25$, $\beta=0.78$) Tier L — low urgency ($\times 15$, $\beta=0.92$)

Recourse markers:

Treated (facility succeeded) Re-manufactured (spare capacity used) Cancelled (no spare capacity / batch fail)